UiT The Arctic University of Norway

Faculty of Engineering science and Technology

D.2.3.1

An immersive and interactive VR virtual planetarium application

Marius Wang

Interreg
Co-funded by the European Union
Northern Periphery and Arctic
GLOW2.0

Table of Contents

1 Introduction 1

2 Virtual planetarium 2

Table of Figures

Figure 1 – Country selection menu 3
Figure 2 – Controller mapping 4
Figure 3 – Menu activated on the controller 4
Figure 4 – Selection of the locations 5
Figure 5 – Menu inside the headset 5
Figure 6 – Adjust the parameters of the skies and weather 6
Figure 7 – List of pre-made scenarios 6
Figure 8 – List of pre-made scenarios part 2 7

1 Introduction

From work package 2 in period 3 – 5 (12.2023 – 05.2025) the activity, 2.3 – The build, test, revise across the scalable outputs:

- Create an Alpha-version build for testing within the project partners and SMEs
- Verification and validation.
- Create a Beta-version build to test on a group of the audience.

The activity output is:

D.2.3.1 – An immersive and interactive VR virtual planetarium application

Output 2.1 – Virtual Planetarium

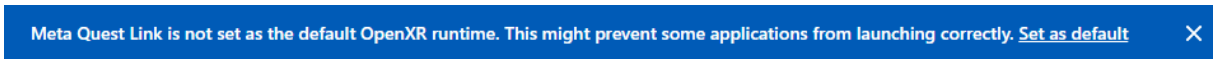
Activity 2.3 includes the design and development of immersive technology with:

- Established multiple locations (Ireland, Iceland, Norway and Finland)
- Design interaction and adapt user experience elements to the virtual reality
- Testing and feedback

Make sure to have Meta Quest Link installed along with SteamVR.

You would have to launch Meta Quest Link first. It is not necessary to launch SteamVR, when opening the Virtual Planetarium, it will do this automatically.

Note: In the dashboard in the Meta Quest Link application, there will be a message prompting that Meta Quest Link is not set as the default OpenXR runtime. Do NOT set it as default, the Virtual Planetarium is running with the SteamVR runtime.



2 Virtual planetarium

Concept chosen was a full-scale virtual environment with stars, earth, earth's moon and with the possibility to control the day/night-time, clouds, and aurora.

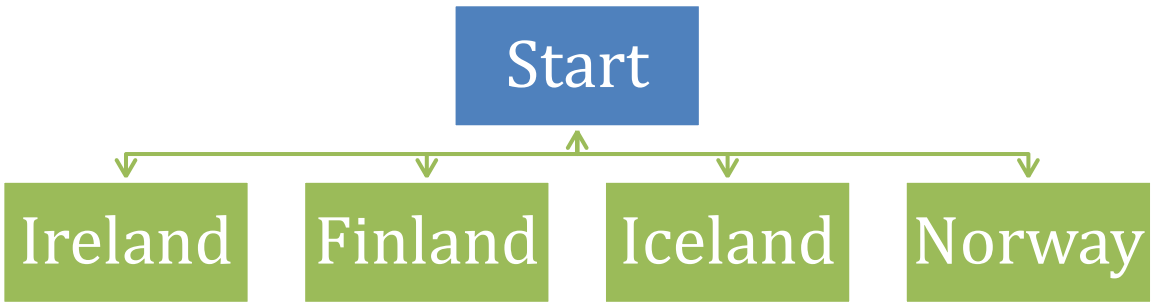
- *Established multiple locations (Ireland, Iceland, Norway and Finland)*

Location in Ireland is Ballycroy (54.023625512826825, -9.821070438792622).

Location in Norway is Narvik (68.43839912909186, 17.427195711951054).

Location in Finland is Koli (63.094396322339804, 29.808824282527556).

Location in Iceland is Hafnir-Rekavatn, (66.08909898169254, -20.38473778450503)



When the Virtual Planetarium is started, the experience will begin in an environment where you can look at the stars without any distractions, select the different locations that you want to visit, or visit the roofless "cabin".

When you are visiting one location, you would have to navigate back to start before traveling to the next destination. You will find this on the menu - "Return to start".

There is ambient music playing, you can control the volume using the volume button on the headset.

- *Design interaction and adapt user experience elements to the virtual reality*

Figure 1 – Country selection menu

This "wall" is where you can select the different locations by moving in front of them, pointing on the flags and using the selection button on the controllers.

Note: The wall will not be visible from the backside.

There is created a menu that could be accessed by either pressing “Y” or “B” buttons on the controllers, on this menu you can change the time and date, weather, reset the orientation, return to start or quit the application.

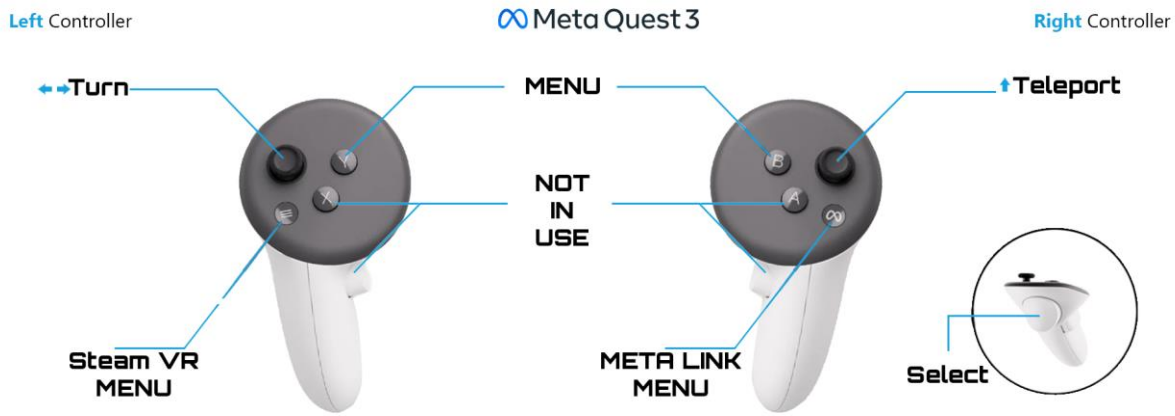


Figure 2 – Controller mapping

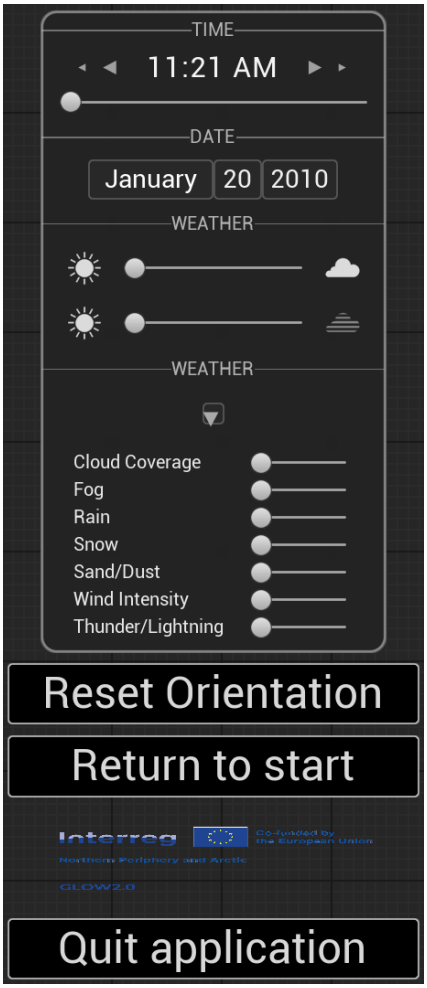


Figure 3 – Menu activated on the controller

Figure 4 – Selection of the locations

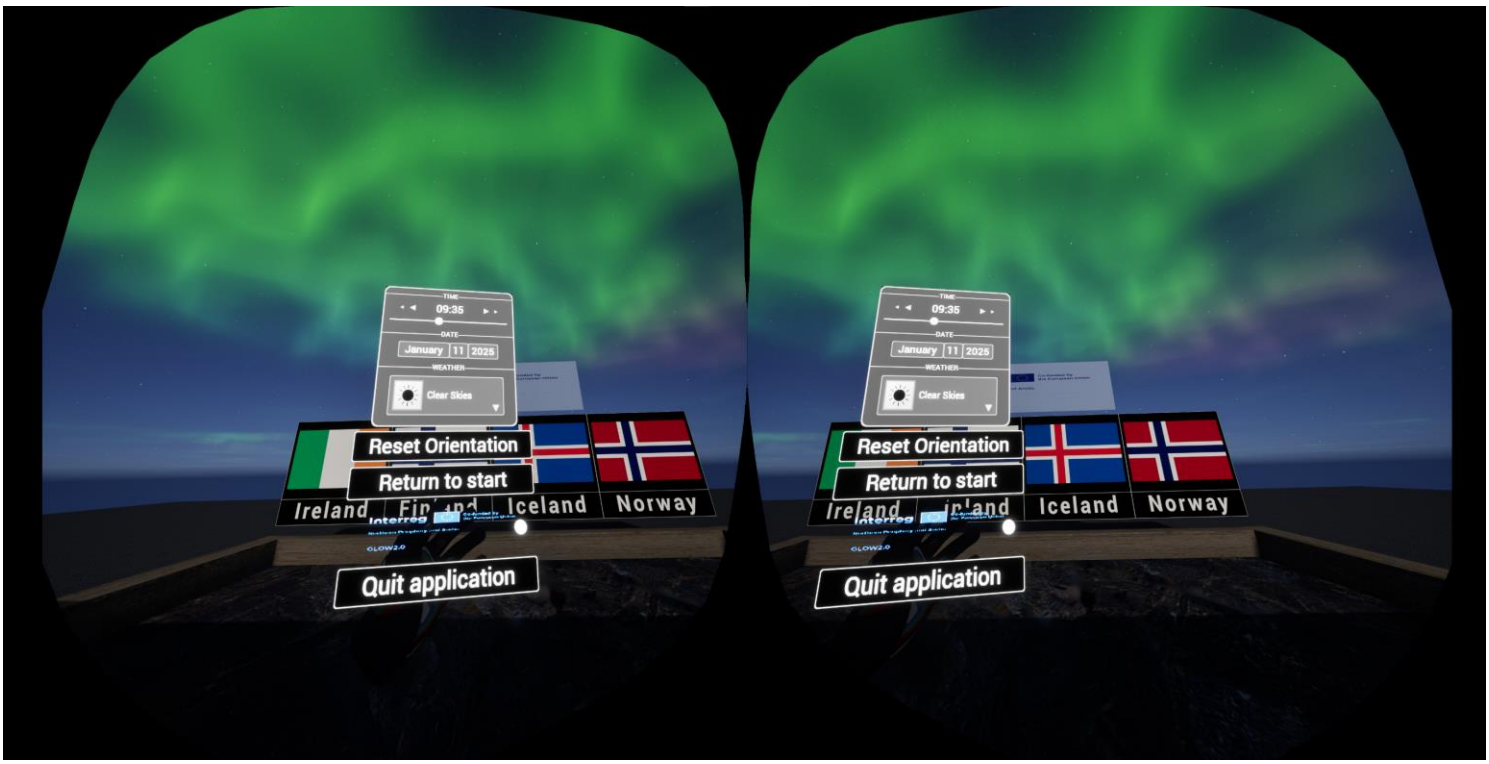


Figure 5 – Menu inside the headset

Figure 6 – Adjust the parameters of the skies and weather

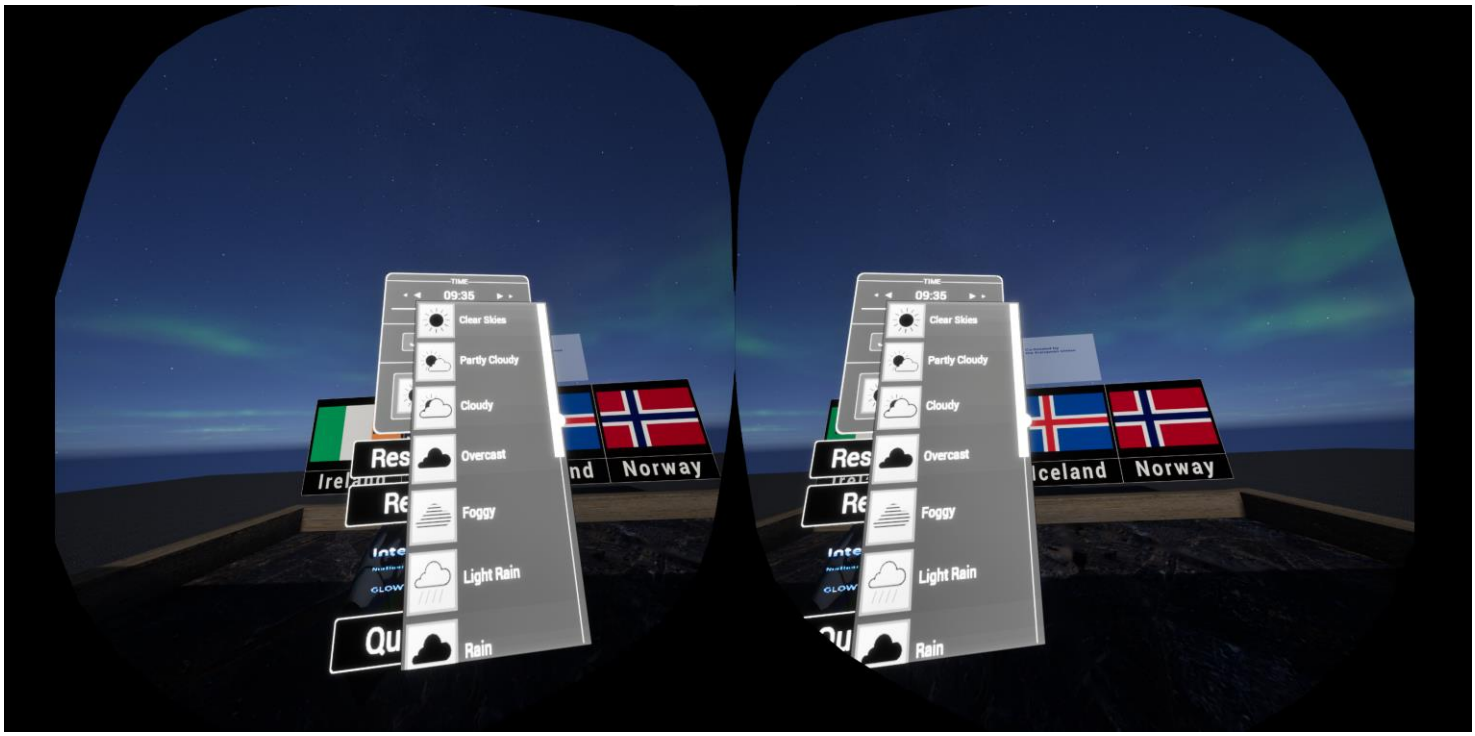


Figure 7 – List of pre-made scenarios

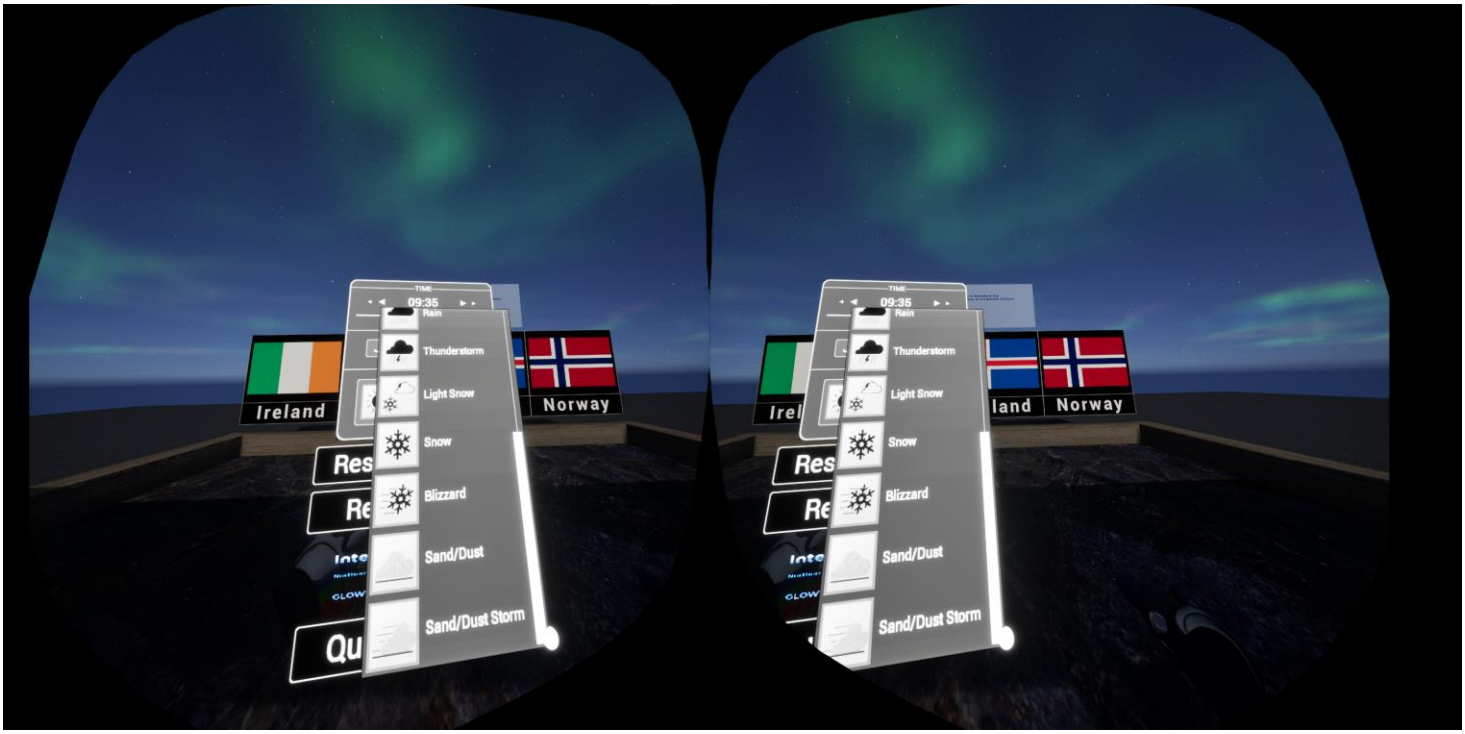


Figure 8 – List of pre-made scenarios part 2

Interreg

Co-funded by the European Union

Northern Periphery and Arctic

GLOW2.0